

Microteaching Project for Linear Algebra (Math 304) First Semester 2025-2026

Student Name	Student ID
Shahad Mohammed Aldossary	2240004665
Fatimah Rashed Almassari	2240004117
Fatimah Gaffar AbuAlmakarem	2240005859
Aisha Mohammed Alhuzaym	2240002568
Arwa Alherz	2240001315
Noor Alhammadi	2240002036
Zainab Hamed Hamada	2240003018
Shahed Ghazi Alharbi	2240000201

1.Introduction

This report presents a **Microteaching Project** for the course **Linear Algebra (Math 304)**. The project focuses on the concept of **Linear Transformations** and how matrices affect the shape, orientation, and position of objects in a 2D plane.

We selected a real 2D figure (**a butterfly image**) and applied four different linear transformations to it:

- **Rotation (180°)**
- **Shear in the y-direction**
- **Scaling / Shrinking**
- **Shear in the x-direction**

For each transformation, we provide:

- **2×2 transformation matrix,**
- **Before -After** comparison of the image
- explanation of the code's purpose.

We used Python on Google Colab to perform the linear transformations, and we used the following libraries: NumPy, OpenCV (cv2), and Matplotlib.

Note: Click on [“Google Colab”](#) to access and view the Colab notebook link , the source code will be shared as a **text file (.txt)**.

2.Code

2.1 Importing Libraries

```
import numpy as np
import cv2
import matplotlib.pyplot as plt
from matplotlib import animation
```

This code is to import the Python libraries we need to read the image, create the transformation matrices, and display the results .

2.2 Load and a Display Original image

```
img_path = "Monarch Butterfly Garden Art.jpeg"

img_bgr = cv2.imread(img_path)

if img_bgr is None:
    print("Image not found. Check the file name.")
else:
    img = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
    plt.figure(figsize=(6,6))
    plt.imshow(img)
    plt.title("Original Butterfly")
    plt.axis("off")
    plt.show()
```

This code loads the (butterfly image) from the file, converts it to the correct color format, and displays the original image to use as the starting point before applying any transformations.

Original Butterfly



2.3 Function to Apply a Linear Transformation

```
def apply_linear_transform(image, A):
    h, w = image.shape[:2]
    cx, cy = w/2, h/2

    M = np.zeros((2,3), dtype=np.float32)
    M[:2,:2] = A

    old_center = np.array([cx, cy])
    new_center = A @ old_center
    t = old_center - new_center

    M[:,2] = t

    transformed = cv2.warpAffine(
        image, M, (w, h),
        flags=cv2.INTER_LINEAR,
        borderValue=(255,255,255)
    )
    return transformed
```

This code defines a function that takes an image and a transformation matrix A , applies the linear transformation while keeping the image centered, and returns the transformed image.

2.4 Defining and Applying Linear Transformation Matrices

2.4.1 Rotation

```
transforms = {}

# 1) Rotation 180 degrees
theta = np.deg2rad(180)
A_rot180 = np.array([
    [np.cos(theta), -np.sin(theta)],
    [np.sin(theta), np.cos(theta)]
], dtype=np.float32)
transforms["Rotation 180 deg"] = A_rot180
```

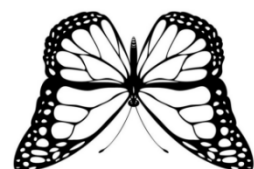
2x2 Matrix A =

```
[-1.0000000e+00 -1.2246469e-16]
[ 1.2246469e-16 -1.0000000e+00]
```

Before



After



2.4.2 Shear in Y

```
# 2) Shear in y-direction (NEW)
k_y = 1.0
A_shear_y = np.array([
    [1, 0],
    [k_y, 1]
], dtype=np.float32)
transforms["Shear in y (k=1.0)"] = A_shear_y
```

Shear in y (k=1.0)

2x2 Matrix A =

[1 0]

[1 1]

Before



After



2.4.3 Shrinking

```
# 3) Shrinking (smaller)
A_shrink = np.array([
    [0.6, 0.0],
    [0.0, 0.6]
], dtype=np.float32)
transforms["Shrinking (0.6x, 0.6y)"] = A_shrink
```

Shrinking (0.6x, 0.6y)

2x2 Matrix A =

[0.6 0.]

[0. 0.6]

Before



After



2.4.4 Shear in X

```
# 4) Stronger Shear in x-direction (more slant)
k_x = 1.2
A_shear_x = np.array([
    [1, k_x],
    [0, 1]
], dtype=np.float32)
transforms["Shear in x (k=1.2)"] = A_shear_x

for name, A in transforms.items():
    out = apply_linear_transform(img, A)

    print("\n" + name)
    print("2x2 Matrix A =")
    print(A)

    plt.figure(figsize=(10,4))

    plt.subplot(1,2,1)
    plt.imshow(img)
    plt.title("Before")
    plt.axis("off")

    plt.subplot(1,2,2)
    plt.imshow(out)
    plt.title("After")
    plt.axis("off")

    plt.show()
```

Shear in x (k=1.2)

2x2 Matrix A =

[1. 1.2]

[0. 1.]

Before



After



These codes create four different 2×2 linear transformation matrices (rotation, shear-y, shrinking, shear-x), apply each one to the original butterfly image, print the matrix, and display a before/after comparison of the result.